Experiment 15

Data Collection and Initial Exploration.

Program:

```
import pandas as pd
diabetes_df=pd.read_csv("diabetes.csv")
print(diabetes_df.head())
print(diabetes_df.info())
print(diabetes_df.describe())
```

Output

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness              768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None

       Pregnancies     Glucose  BloodPressure  SkinThickness      Insulin
count   768.000000  768.000000     768.000000     768.000000   768.000000
mean      3.845052  120.894531      69.105469      20.536458    79.799479
std       3.369578   31.972618      19.355807      15.952218   115.244002
min       0.000000    0.000000       0.000000       0.000000     0.000000
25%       1.000000   99.000000      62.000000       0.000000     0.000000
50%       3.000000  117.000000      72.000000      23.000000    30.500000
75%       6.000000  140.250000      80.000000      32.000000   127.250000
max      17.000000  199.000000     122.000000      99.000000   846.000000
■          BMI  DiabetesPedigreeFunction         Age     Outcome
count  768.000000              768.000000  768.000000  768.000000
mean    31.992578                0.471876   33.240885    0.348958
std      7.884160                0.331329   11.760232    0.476951
min      0.000000                0.078000   21.000000    0.000000
25%     27.300000                0.243750   24.000000    0.000000
50%     32.000000                0.372500   29.000000    0.000000
75%     36.600000                0.626250   41.000000    1.000000
max     67.100000                2.420000   81.000000    1.000000
```